

Prashans Chauhan

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Education

Year	Degree / Qualification	Institution	Score
Nov 2022 – Jul 2026	B.Tech in Engineering Science	IIT Hyderabad	GPA: 8.62/10
Jan 2024 – May 2026	Minor in Entrepreneurship Management	IIT Hyderabad	GPA: 8.45/10
2022	XII (CBSE)	Central Academy School, Gwalior	95.00%
2020	X (CBSE)	Kendriya Vidyalaya No.2, Gwalior	94.00%

Work Experience

Project Intern – NL-to-SQL & Agentic AI *May 2025 – Jul 2025*
Oracle | *Bangalore, India*

- Built a **RAG pipeline** for **NL-to-SQL translation** on Oracle's enterprise ERP as part of a **4-member research team**, enabling non-technical users to retrieve data via natural language directly targeting **reduction in customer consultation costs**.
- Designed an **agentic workflow** using Oracle's internal **Agent Studio**; conducted **LLM benchmarking** identifying critical **failure modes in multi-table joins**, informing prompt engineering and model selection decisions.

Projects

NaviGit – GitLab Knowledge Assistant *Mar 2026*

- Developed a **full-stack RAG pipeline** to query **500+ GitLab handbook pages**, achieving a **92% retrieval hit rate** by optimizing **Jina-v4 task-specific embeddings** and a sliding-window chunking strategy.
- Engineered a **Gemini-powered guardrail** filtering off-topic queries with **96% accuracy**, reducing **LLM inference costs and token usage by 20%**.
- Achieved **88% faithfulness score** (RAGAS framework) via grounded generation with **Gemini 2.0 Flash**; decreased perceived response latency by **65%** using **SSE token streaming**.

Online Code Judging System | Distributed Code Judge *2025*

- Built a **secure code execution engine** using **Fastify + BullMQ (Redis)**, architecting **sandboxed Docker containers** per submission for isolated Java/C++ execution with strict timeouts and resource limits.
- Implemented **worker orchestration** with retry/backoff and result persistence, ensuring **reliable async verdict delivery**; applied **Nginx reverse proxy** with Dockerized multi-service deployment.

AI Meeting Assistant – Local Video Intelligence Tool *2026*

- Built a **local AI Video Assistant** using **Python** and **Streamlit** to transcribe and analyze meetings, serving as a free alternative to premium platforms like Fireflies and Otter.ai.
- Integrated a **multi-lingual transcription engine** standardizing audio via **ffmpeg** and processing long-duration inputs in manageable chunks using **OpenAI Whisper** and **Sarvam AI**.
- Engineered a **RAG-based intelligence pipeline** with **ChromaDB**, **HuggingFace**, and **Mistral AI** via **LangChain LCEL** to automatically extract action items and enable semantic chat-to-meeting queries.

Continual Learning & Unlearning in LLMs *Aug 2025 – Jan 2026*

- Tackled the **joint CL + MU problem** on **LLaMA-3.1-8B** over **6 interleaved tasks**, extending the TOFU benchmark with **4,000 generated QA pairs**; deployed memory-efficient **dual-adaptor QLoRA** on 24 GiB VRAM.
- Applied **Elastic Weight Consolidation (EWC)** for continual learning, hitting **80% learn-set retention** while dropping forget-set accuracy to **10%**.
- Adapted **Gradient Surgery** to stop the model from re-learning what it already forgot; enabled **concept-level unlearning** via LLM prompting across the full task sequence.

SafeDialog: Intent-Aware Detection of Mental Manipulation *Jan – Apr 2026*

- Designed an **Intent-Aware Pipeline (IAP)** which extracts speaker's psychological intent before classification using Llama 3, achieving **22% accuracy gain & 3.5× lower false negatives** over CoT baseline.
- Extended IAP to **Intent-Aware DistilBERT**; focal loss & threshold calibration achieved **26% F1 gain over best prompting baseline** while **cutting false negatives by 3×**.
- Ablated **6 prompting strategies** and **4 models** on an imbalanced dataset; identified decoupling detection from response generation eliminates mode collapse; established **Guardian architecture as the most robust design**.

- Engineered a **three-tier architecture** combining a **ViT backbone**, **shared LoRA adapters** & **private classification heads** enabling personalized label spaces via collaborative learning across **4 federated hospital clients**.
- Achieved **168×** communication compression via LoRA parameter sharing; conducted **4×2 factorial study** identifying **Knowledge Distillation + FedProx** as optimal FL-CL configuration.
- Achieved **45.1% Task-1 retention** & **85.88% Task-2 accuracy**, reducing backward transfer by **44.63%** vs. baseline.

Airline Booking System | Microservices

2026

- Architected a **production-grade microservices backend** (Flights, Bookings, Auth) using **Node.js**, **Express**, **PostgreSQL**; built **idempotent APIs** with **ACID transactions** and concurrency control to eliminate double-booking race conditions.
- Integrated **Apache Kafka** for async notifications ensuring **eventual consistency**; Dockerized services with **Nginx** as reverse API gateway and validated scalability via **k6 load testing**.

Smart Resource Booking System for IIT Hyderabad

2026

- Designed a modular resource-booking backend using **NestJS**, **TypeScript**, **MongoDB**, and **Redis caching**, architecting booking lifecycle workflows with **approval history**, **waitlisting**, **status transitions**, and **check-in/check-out tracking**; built validated **RESTful APIs** with Swagger integration.
- Developed an **LLM-enabled booking assistant** supporting natural-language queries with **intent detection**, **entity extraction**, **session memory**, and **recommendation logic**; scaled to **5,000+ concurrent users** via Redis caching, Winston logging, and centralized error handling.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML/CSS

AI & ML: PyTorch, HuggingFace Transformers, LLM Fine-Tuning (LoRA/PEFT), RAG Pipelines, Federated Learning, LLM Agents, Multi-Agent Systems, Scikit-Learn

NLP & LLMs: Continual Learning, Model Unlearning, Intent Extraction, Text Embeddings, LangChain, LangGraph

Backend & Databases: Node.js, Express.js, Fastify, FastAPI, PostgreSQL, MongoDB, Redis, Apache Kafka, REST APIs

DevOps & Tools: Docker, Nginx, Git, Linux, k6, NumPy, Pandas, Streamlit, Jupyter

Achievements

- Optiver Trade-a-Thon (2026):** Secured **2nd rank** in Inter-IIT Trading Competition organized by Optiver.
- Synchrony Hackathon (2025):** Silver Medallist – Inter-college hackathon across Hyderabad.
- JEE (2022):** Advanced – **AIR 4008** (top 0.2%)
- Problem Solving:** Solved **650+ DS & Algo** problems across competitive coding platforms.
- PRMO (2018 & 2019):** Ranked top 300 among 10,000+ participants in both years; qualified for the Regional Mathematical Olympiad (RMO).

Extracurriculars

- Head Placement Coordinator, OCS:** Managed a team of 60+ coordinators and volunteers for pitching, organizing interviews & events.

Aug 2025 – Present
- Sunshine Buddy, Sunshine Club (Mental Health Club):** Provided peer mental health support and one-on-one guidance to junior students as part of IITH’s student wellness initiative.

Dec 2024

Relevant Courses

Computer Science: Software Engineering, Natural Language Processing, Computer Vision, Deep Learning, Foundations of Machine Learning, Database Management Systems

Mathematics: Probability and Random Variables, Introduction to Statistics, Linear Algebra, Transform Techniques, Calculus